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23 AUGUST 1962
SUPERSEDING
JAN–I–706
22 DECEMBER 1948

MILITARY SPECIFICATION

IRON OXIDE; FERRIC, RED DRY (NATURAL AND SYNTHETIC)

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers 3 classes of dry, ground iron oxide; ferric, and synthetic red.

1.2 **Classification.** Ferric iron oxide covered by this specification shall be of the following classes differentiated by grain size.

Class 1—Coarse
Class 2—Fine
Class 3—Extra fine (Synthetic ferric iron oxide)

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP–D–705 —Drum: Metal Shipping, Steel (Over 12 and Under 55 Gallon)
PPP–D–723 —Drums, Fiber
PPP–D–729 —Drums; Metal 55-Gallon (for Shipment of Non-corrosive Materials)
PPP–T–76 —Tape, Pressure-Sensitive Adhesive, Paper, Water Resistant
RR–S–366 —Sieves, Standard for Testing Purposes
UU–P–268 —Paper, Kraft, Wrapping

MILITARY

JAN–P–112 —Packaging and Packing for Overseas Shipment-Drums, Plywood

STANDARDS

MILITARY

MIL–STD–105 —Sampling Procedures and Tables for Inspection by Attributes
MIL–STD–129 —Marking for Shipment and Storage

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

CODE OF FEDERAL REGULATIONS

49 CFR 71–90 —Interstate Commerce Commission, Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Orders for copies should site "49 CFR 71–90 (Revised 1956)".)

FSC 6810

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AMERICAN SOCIETY FOR TESTING MATERIALS

ASTM–E50–53 —Apparatus and Reagents for Chemical Analysis of Metals, Jones Reductor (1956 revision)

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pennsylvania.)

CONSOLIDATED CLASSIFICATION COMMITTEES

Consolidated Freight Classification Rules

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Illinois.)

AMERICAN TRUCKING ASSOCIATIONS

National Motor Freight Classification

(Application for copies should be addressed to the National Classification Board, 1424 Sixteenth Street, Washington 6, D. C.)

3. REQUIREMENTS

3.1 Preproduction sample. When a preproduction sample is required (see 6.2), it shall meet all requirements of this specification, and shall be produced by the same methods, processes, and techniques to be employed in subsequent production of the contract quantity.

3.2 Composition. The composition of the iron oxide (Fe_2O_3) shall conform to the requirements specified in Table I. Class 3 material shall be red, dry, synthetic, ferric, iron oxide.

TABLE I

Composition	Percent
Water-soluble material, maximum	0.30
Acidity, maximum (as sulfuric acid (H_2SO_4)	0.01
Volatile matter, maximum	0.50
Silica, maximum	0.15
Purity (as Fe_2O_3), minimum	98.0
Calcination loss (class 3 only)—maximum	1.50

3.3 Grain size. Grain size of the Fe_2O_3 shall conform to the requirements of table II, using U.S. Standard sieves conforming to the requirements of Federal Specification RR–S–366.

TABLE II.

Sieve No.	Class 1	Class 2	Class 3
	Percent minimum	Percent minimum	Percent minimum
40	100	.....	.....
100	85	.....	.....
200	.....	99.9	.....
325	.....	85	99.0

3.4 Workmanship. The iron oxide shall be a homogeneous crystalline composition and shall be free of lumps and any foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and test shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot size. A lot of iron oxide shall consist of that quantity produced by one manufacturer, in not more than 24 consecutive hours of continuous production, with no change in facilities, materials, process or specifications which would change the chemical or physical properties of the composition. In the event the process is a batch operation, each batch shall constitute a lot.

4.3 Preproduction provisions.

4.3.1 *Sample.* The preproduction sample shall be that quantity of iron oxide specified by the procuring activity (see 6.2).

4.3.2 *Inspection location.* Preproduction inspection shall be performed by the Government at a place specified by the procuring activity (see 6.2).

4.4 Production provisions.

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4.4.1 *Sampling.*

4.4.1.1 *For examination.* Sampling for examination shall be conducted in accordance with Standard MIL–STD–105 at inspection level II, acceptable quality level (AQL) 1.50 percent.

4.4.1.2 *For chemical and physical tests.*

4.4.1.2.1 *Primary samples.* Ten percent of the containers comprising each lot submitted shall be selected at random; except when lots comprise less than 100 containers, 10 or all the containers, whichever is less, shall be selected. A representative one-pint sample from each selected container shall be taken in the following manner: Insert a closed sampler tube to maximum depth, at an oblique angle, then open and agitate to fill the tube. Each sample shall be placed in a clean, dry glass jar, which shall be labeled to identify the container and lot number.

4.4.1.2.2 *Composite samples.* The primary samples shall be divided into two groups of equal size. Equal amounts of each primary sample shall be blended together to form a composite sample. Each composite shall be placed in a clean, dry container, sealed, and labeled in a manner that will identify the sample with the container and lot represented. Each composite sample shall be subjected to all chemical and physical tests specified in 4.5, and the average value of results obtained from testing the two composite samples shall be taken as representative of the lot.

The Government inspector, when examining the contractor's inspection equipment, will determine that the contractor has available, and utilizes correctly, gaging, measuring, and test equipment of required accuracy and precision, and that the instruments are of a proper type and range to make measurements within the required accuracy. The contractor shall have available a set of master gages, standards, and appropriate instruments to conduct regularly scheduled calibrations of this inspection equipment. Records of such calibrations shall be maintained by the contractor and made available for Government review. The calibration of gages, standards, and instruments will be checked periodically by authorized Government personnel.

4.5 Tests.

4.5.1 *Chemical tests.*

4.5.1.1 *Water-soluble material.* Boil 2 grams (gm) of the sample with 25 milliliters (ml) of distilled water, allow to settle, and decant the supernatant liquid through a fine porosity filter. Repeat this operation 3 times using 25 ml portions of distilled water each time. Catch the combined filtrates in a tared weighing bottle or beaker, evaporate to dryness, heat to constant weight at 105 degrees Centigrade (C). Calculate the increase in weight as percent water-soluble material.

4.5.1.2 *Acidity.* To 10 gm of the sample, add 100 ml of water, boil, filter, and wash with hot water. Test filtrate with litmus paper, and if acid, titrate with 0.1N potassium hydroxide using methyl orange as indicator. Calculate the acidity in sulfuric acid (H_2SO_4) as follows:

$$H_2SO_4, \text{ percent} = \frac{4.9VN}{W}$$

Where,

V = ml of potassium hydroxide solution used,

N = normality of potassium hydroxide, solution

W = gm of sample.

4.5.1.3 *Volatile matter.* The determination shall be made as follows: Weigh a sample of 2 or 3 grams in a tared low-form, glass-stoppered weighing bottle (about 50 millimeters (mm)). Place in a 105° C oven for 90, plus or minus 3, minutes, cool in a desiccator, over desiccant, and weigh. Calculate percent volatile matter as follows:

$$\text{Volatile matter, percent} = \frac{(A-B)\ 100}{(A-C)}$$

Where,

A = Weight of the weighing bottle and sample,

B = Weight of the weighing bottle and dried sample,

C = Weight of dry weighing bottle.

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4.5.1.4 *Silica.* Weigh a sample of 5 gm into a beaker. Add 20 ml of HCl (specific gravity (sp gr) 1.18) and heat to dissolve. Cool, add 30 ml HNO_3 (1:1) and 30 ml of 60 percent perchloric acid. Carefully evaporate the solution to fumes, and then boil for 15 minutes. Caution should be observed in the volatilization of perchloric acid. In order to prevent the accumulation of perchloric acid in the flue of the hood, it is suggested that at this point in the procedure, the acid fumes from the beaker be drawn into a water trap by means of a funnel and suction. Cool, dilute with 125 ml of hot water and stir to dissolve all soluble salts. Filter the solution on No. 40 Whatman paper, or equivalent, wash the residue free of perchloric acid and salts with HCl (1:20), and then with several portions of hot water. Transfer paper and residue to a clean platinum crucible, ignite gently until the filter paper is consumed and finally at a bright red heat. Cool in a desiccator and weigh. Add 3 to 4 drops of H_2SO_4 (sp gr 1.84) and approximately 5 ml of hydrofluoric acid. Evaporate to dryness on a hot plate, heat again to a bright red heat, cool and weigh. Calculate the loss in weight between weighings as percent silica.

4.5.1.5 *Purity* (as Fe_2O_3). Accurately weigh a dried sample of approximately 0.2 gm and transfer to a beaker. Add 20 ml of 1:1 HCl and heat. Upon solution of the sample, add ml of 1:1 H_2SO_4 and evaporate, without boiling, to fumes of SO_3 in order to remove the HCl. Cool, dilute with 100 ml of water, add a slight excess of permanganate ($KMnO_4$), and boil. Cool, and pass the solution through a Jones reductor prepared in accordance with ASTM-E50-53. Wash with 50 ml of 1:20 H_2SO_4 and then with 150 ml of water. Titrate with 0.1N $KMnO_4$ solution and correct for a blank. Calculate the purity as follows:

$$\text{Purity as percent } Fe_2O_3 = \frac{7.985\,(V-v)\,N}{W}$$

Where,

V = ml of $KMnO_4$ used to titrate the sample,

v = ml of $KMnO_4$ used to titrate the blank,

N = normality of $KMnO_4$ solution,

W = gm of sample

4.5.1.6 *Calcination loss.* Accurately weigh a one gram sample into a tared platinum crucible. Place the lid on the crucible and ignite at 800-900° C for 90 minutes. Cool in a desiccator and weigh. Calculate the loss in weight as loss on ignition as follows:

$$\text{Calcination loss, percent} = \frac{(A-B)\,100}{W}$$

Where,

A = weight of crucible plus sample,

B = weight of crucible plus ignited sample,

W = weight of sample.

4.5.2 *Physical tests.* The grain size shall be determined as follows:

4.5.2.1 *General.* Select U. S. Standard sieves in accordance with 3.3 for these tests. Nest the specified sieves in order of decreasing size, placing the largest mesh on top and a pan at the bottom. Accurately weigh a 50 gm sample of the material and place it in the top sieve.

4.5.2.1.1 *Class 1.* Covering the specified nest of sieves (see Table II), shade for 30 minutes using a machine of the type geared to produce 300, plus or minus 15, gyrations and 150, plus or minus 10, taps of the striker per minute. Weigh the material remaining on each sieve and the bottom pan and calculate the percentages as required.

4.5.2.1.2 *Class 2.* Brush the material on the No. 200 sieve with a camel's-hair brush until no more material passes through the screen. Remove the sieve and weigh the material left on the sieve. Repeat the procedure with the material left on the No. 325 sieve. Calculate to percentages as required.

4.5.2.1.3 *Class 3.* Brush the material on the No. 325 sieve with a camel's-hair brush until no more material passes through the screen. Remove the sieve and weigh the material left on the sieve. Calculate to percentages as required.

4.6 *Packing and marking.* Visual and tactile examination of packing and marking

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shall be performed to determine compliance with the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 *Level A.* The iron oxide shall be packed directly in drums 25 and 100 pound net capacity, as specified by the contract or order. The 25 pound quantities shall be packed in drums conforming to Specification PPP–D–705a, Type VI. The 100 pound quantities shall be packed in drums conforming to Specification PPP–D–729, Type IV. The drums shall be fabricated from steel not lighter than 26 gage for the 25 pound capacity, and not lighter than 28 gage for the 100 pound capacity. The drums shall have a bolted ring-type cover.

5.1.2 *Level B.* The iron oxide shall be packed directly in water-resistant fiberboard or plywood drums conforming to Specification PPP–D–723, Type II, Grade A, and Specification JAN–P–112, respectively. Net capacity of the drums shall be 25, 50, or 100 pounds as specified in the contract or order. Each plywood container shall be lined with a kraft paper bag fabricated from material conforming to Specification UU–P–268. After filling, the ears or flaps of the liner shall be secured in place with tape conforming to Specification PPP–T–76. The top-closure and strapping shall be in conformance to the applicable container specification.

5.1.3 *Level C.* Unless otherwise specified (see 6.2), the iron oxide shall be packed 100 pounds in bulk, in a manner to afford adequate protection during shipment from the supply source to the first receiving activity for immediate use. The shipping containers shall comply with the Consolidated Freight Classification Rules and container Specifications for rail shipment or with National Motor Freight Classification Rules and container Specifications for truck shipments.

5.2 Marking.

5.2.1 *Normal marking.* The exterior shipping container shall be marked in accordance with the requirements of Standard MIL–STD–129 and Interstate Commerce Commission requirements specified in Code of Federal Regulations 49 CFR 71–90.

5.2.2 *Special marking.* In addition to any special marking required by the contract or order, digit marking of exterior containers for overseas shipment shall be accomplished as specified by the procuring activity (see 6.2).

6. NOTES

6.1 **Intended use.** Ferric iron oxide covered by this specification is intended for use in pyrotechnic mixtures and propellants.

6.2 **Ordering data.** Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Whether a preproduction sample is required (see 3.1 and 4.3.1) and size of sample (see 4.3.1).
(c) Preproduction inspection location (see 4.3.2).
(d) Selection of applicable level of packing (see 5.1).
(e) Marking requirements, if other than specified (see 5.2.1).
(f) Whether digit marking is required (see 5.2.2).

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